

Do LLMs Choose Concepts or English Words?

Answer-Side Translation in CommonsenseQA

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Paper Chosen: CommonsenseQA: A Question Answering Challenge Targeting Commonsense Knowledge.

Brief Description of the Project:

CommonsenseQA was originally designed in English. Prior multilingual work, such as X-CSQA, translated the full question-answer items into many languages to evaluate cross-lingual commonsense reasoning. However, our project studies a more focused question:

Do language models choose the correct commonsense concept, or do they rely on English answer wording?

To test this, we will keep the CommonsenseQA questions in English but translate only the answer choices into other languages. This creates a controlled mixed-language setting: the model understands the English question but must map it to candidate concepts expressed in another language.

Resources:

- CommonsenseQA as the base dataset.
- Answer-choice translations into several target languages, likely including Hebrew, Arabic, and one high-resource language such as Spanish or French.
- Modern LLMs and/or multilingual language models for evaluation.
- Optional human validation for a small subset to check whether translated answer choices preserve the original meanings.

Methods:

We will compare several versions of the same examples. The question Q and answers A can be permuted in the following Q-A way: EN-EN, EN-X, X-X, X-EN. Where X is a translated language. We will check different translations in the same answer set, i.e. some answers translated in one language and the rest in another. We will translate some nouns in the questions to see the effect of mixed questions. This isolates whether LLMs can ground answer concepts across languages and measures prediction flips caused solely by changing answer language. To the best of our knowledge, prior multilingual CQA work has focused on full-item translation, language-specific dataset construction, or question-side code-switching. Answer-side-only translation remains underexplored as a diagnostic for cross-lingual concept grounding.

References

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